Formulas where Subject Appears Twice

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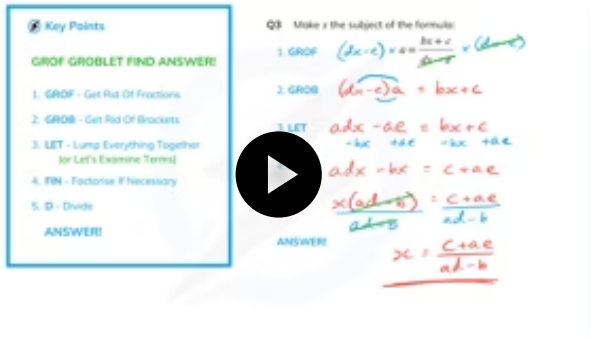

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Subject appears twice

How do I rearrange formulae where the subject appears twice?

- If the **subject appears twice**, **you will need to factorise** at some point
  - E.g. When making x the subject of $x + xy = 3 - 2y$
  - **Factorise** out x on the left to get $x(1 + y) = 3 - 2y$
    - Notice that the subject **now only appears once!**
  - Then divide both sides by $(1 + y)$ to get $x = \frac{3 - 2y}{1 + y}$
- If the **subject appears twice**, and any of these are **inside a set of brackets**, you will need to **expand** these brackets first
  - E.g. When making x the subject of $c(x + 2) - x = f$
  - Expand the bracket first to $cx + 2c - x = f$
  - Keep the x terms on one side $cx - x = f - 2c$
  - x can then be made the subject using **factorising** as above
- If the **subject appears on two sides of a formula**, you will need to bring those terms to the **same side** before you can factorise
  - E.g. When making x the subject of $3x = y - px$

Help me understand

Stuck on something?

Ask questions and get help with exactly what you're revising.

Got it

How do I factorise powers of a subject?

- If the **subject appears twice**, and **both have the same power**, you will need to **collect these terms together** before applying their inverse
  - E.g. When making x the subject of $x^2 = -px^2 + r$
  - Add px^2 to both sides first to form $x^2 + px^2 = r$
  - x^2 can then be factorised out $x^2(1+p) = r$ to give $x^2 = \frac{r}{1+p}$
    - Now take **plus-or-minus square roots**
    - $x = \pm\sqrt{\frac{r}{1+p}}$
- Be careful when square rooting, or cube rooting etc
  - E.g. To make x the subject of $x^3 = \frac{t^3+1}{t^3+8}$
    - The whole right hand side must be cube rooted
    - $x = \sqrt[3]{\frac{t^3+1}{t^3+8}}$
    - This cannot be simplified further
    - The right hand side is **not** equal to $\frac{t+1}{t+2}$, (this is a **common error**)

Worked Example

Rearrange the formula $p = \frac{2 - ax}{x - b}$ to make x the subject.

Answer:

Get rid of the fraction by multiplying both sides by the expression on the denominator

$$p(x - b) = 2 - ax$$

Expand the brackets on the left hand side to 'release' the x

$$px - pb = 2 - ax$$

Bring the terms containing x to one side of the equals sign and any other terms to the other side

$$px - pb = 2 - ax$$

$$(+ax) \qquad\qquad (+ax)$$

$$px - pb + ax = 2$$

$$(+pb) \qquad\qquad (+pb)$$

$$px + ax = 2 + pb$$

Factorise the left-hand side to bring x outside of the brackets, so that it appears only once

$$x(p + a) = 2 + pb$$

$$x = \frac{z + pb}{p + a}$$

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